SWIKAR PATEL

Staff Software Engineer AI/ML & Distributed Systems Architect

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SUMMARY

With over 6 years of enterprise experience designing fault-tolerant microservices, high-throughput machine learning ranking systems, and zero-trust AI architectures. Proven track record bridging classic backend distributed systems (Java/Spring Boot, Apache Kafka, low-latency microservices) with modern predictive ML (XGBoost Learning-to-Rank, Feature Stores, MLOps) and Gen AI (Agentic FSM Runtimes, GraphRAG via MCP). Expert in sub-50ms p99 serving latency, position debiasing, online-offline feature skew detection, and end-to-end ML lifecycle ownership.

EXPERIENCE

CAPITAL ONE | Staff AI Software Engineer / AI Architect (2025 – 2026)

- **Machine Learning & Ranking Systems:** Learning-to-Rank (LTR), XGBoost, Two-Stage Candidate Retrieval (H3 Geospatial Indexing, Inverted Index), Feature Stores (Redis/Feast), Position Debiasing (Inverse Propensity Scoring), Multi-Objective Optimization, Scikit-Learn, PyTorch.
- **Standardized Agent-to-Agent (A2A) Protocols:** Engineered structured JSON-RPC messaging envelopes with an Anti-Amplification Supervisor that blocked low-confidence (<0.85) handoffs across distributed banking agents.
- **Generative AI & Agentic Systems:** GraphRAG (Neo4j), Model Context Protocol (MCP), Deterministic Finite State Machines (FSM), AgentScript DSL, Multi-Agent A2A Protocols, Human-in-the-Loop (HITL) Suspension, Token Vaults (Zero-Trust ITAR/PII).
- **Zero-Trust Data Clean Room & MCP Tooling:** Built local AES-256 token vaults that masked sensitive PII/account numbers before cloud inference, and isolated Neo4j knowledge graphs via Anthropic's Model Context Protocol (MCP) JSON-RPC sandbox.
- **Architected Enterprise Agentic Middleware:** Designed stateful, deterministic FSM Agentic AI runtimes for high-stakes wire transfer fraud investigations processing $40M+ daily transactional volume.

T-MOBILE | Senior Software AI / ML & Distributed Systems Engineer (2021 – 2022)

- **Distributed Systems & Backend:** Java (Spring Boot), Python (FastAPI, asyncio), RESTful Microservices, gRPC, Distributed Consensus, High-Concurrency Serving (<50ms p99 SLA), Circuit Breakers (Resilience4j).
- **Data Infrastructure & MLOps:** Apache Kafka, Apache Spark, Snowflake, PostgreSQL, Qdrant, SQLite, Redis, Automated Airflow Retraining DAGs, Population Stability Index (PSI) Drift Detection, MLflow.
- **Implemented Position Debiasing & Marketplace Fairness:** Solved exposure bias during training using the position-zeroing inference trick, and engineered a utilization-penalty algorithm to prevent winner-take-all supplier saturation.
- **High-Scale Intent Matching & Ranking Engine:** Built real-time machine learning ranking models serving 10M+ customer sessions using PyTorch and XGBoost, driving a 14% lift in conversion rates.

- **Automated MLOps & Drift Monitoring Pipelines:** Engineered automated retraining DAGs in Airflow integrated with Z-score and Population Stability Index (PSI) anomaly detectors, cutting model performance degradation by 65%.
- **Cloud, DevOps & Security:** AWS (EKS, ECS, Lambda, KMS, S3), Docker, Kubernetes, Terraform (IaC), CI/CD, OpenTelemetry Cryptographic Distributed Tracing, AES-256 Encryption.
- **Engineered Hardware-Aware Serving & Latency Optimization:** Built an AST complexity profiler and memory router that dynamically shifted workloads between cloud LLMs and local SLMs, reducing cloud token costs by 40% and maintaining strict sub-50ms p99 latency.
- **Real-Time Kafka Streaming & Telemetry:** Architected streaming pipelines using Apache Kafka and Apache Spark, processing billions of daily network event logs to build predictive congestion models.

CAPITAL ONE | Senior Software Engineer (Distributed Microservices) (2020 – 2021)

- **High-Throughput Financial Microservices:** Engineered core transactional RESTful microservices in Java (Spring Boot) and Python on AWS (EKS, Lambda) supporting 50,000+ TPS with 99.99% uptime.
- **Distributed Tracing & Fault Tolerance:** Implemented OpenTelemetry distributed tracing and Resilience4j circuit-breaker patterns across 30+ microservices, cutting mean-time-to-resolution (MTTR) by 50% and ensuring SOC2/SEC auditability.
- **Zero-Downtime Data Lake Migration:** Spearheaded automated migration pipelines transferring legacy financial transaction records into AWS S3/Snowflake data lakes with automated PII tokenization.

ERICSSON | Software Engineer (Telecom Systems & ML) (2018)

- **Real-Time Network Anomaly Detection:** Developed Python data pipelines for real-time cellular tower telemetry analysis, reducing false-positive outage alerts by 30%.
- **Graph Network Dependency Modeling:** Modeled regional cell tower network topologies to optimize automated load-balancing algorithms during peak traffic events.

IBM | Software Engineer (Enterprise Middleware) (2016 – 2017)

- **Cognitive Computing Middleware:** Integrated enterprise Watson NLU AI APIs into Java/REST cloud microservices, delivering scalable cognitive middleware for Fortune 500 enterprise clients.
- **API & High-Volume Backend Development:** Built and maintained high-throughput web services and blockchain platforms using Hyperledger Fabric in Agile/Scrum environments.

EDUCATION

Bachelor of Science (B.S.) in Computer Science / Software Engineering
James Cook University | 2007 – 2011